

MAST32001 Computational Statistics

Exercises 3 — Permutation Tests and the Bootstrap

University of Helsinki

Instructions

These exercises are based on Lectures 5 and 6.

The aim is not only to run permutation or bootstrap code, but to decide **which resampling scheme is justified and what question it answers**.

For each exercise:

- state the null hypothesis or target parameter clearly;
- identify what is assumed to be exchangeable or iid;
- choose the statistic before resampling;
- interpret the result in the context of the problem;
- use a fixed random seed whenever Monte Carlo resampling is used.

Submission

Solutions must be submitted in **Moodle no later than 22.09.2026 at 12:00**.

Please submit your solutions in the **submission area corresponding to your exercise group**. Make sure that you submit to the correct exercise group.

Exercise points

Exercise points are awarded only in connection with the corresponding exercise session.

To receive the point(s) for a task, you must:

1. submit your solution by the deadline;
2. attend the exercise session in which the task is discussed; and
3. be willing to **present, explain, and discuss your solution** during the exercise session if asked.

Submitting a solution without attending the corresponding exercise session does **not** earn exercise points.

Your solution does not need to be perfect in order to be discussed. The purpose of the exercise sessions is also to compare approaches, identify difficulties, and learn from different solutions.

Use of generative AI

The use of **generative AI tools to solve the exercise tasks is not permitted**.

The purpose of the exercises is to develop your own ability to translate statistical ideas into algorithms, implement them in R, investigate numerical behaviour, and explain your reasoning.

The submitted solutions, including the reasoning and R code, must therefore reflect **your own work**.

Exercise 1 — What may be permuted?

A study compares a numerical response between two groups. Consider the following three data-generating situations.

A. Randomized experiment.

Twenty-four experimental units are randomly assigned to treatment or control, with 12 units in each group.

B. Observational study.

Twelve observations are sampled from population 1 and twelve independently from population 2.

C. Matched pairs.

Twelve people are measured before and after an intervention.

Tasks

1. In situation A, explain why treatment labels can be permuted under the sharp null hypothesis of no treatment effect.
2. In situation B, suppose the null hypothesis is

$$H_0 : F_1 = F_2.$$

Explain why group labels are exchangeable under this null.

3. In situation C, explain why freely permuting the 24 labels would destroy the matched-pair structure.
4. Propose a valid permutation scheme for situation C under a null hypothesis of no systematic before/after effect.
5. For each situation, state precisely **what is being permuted** and **what must remain fixed**.
6. Explain why the statement

“A permutation test is nonparametric, so we may always shuffle the data”

is incorrect.

Exercise 2 — What does the plot suggest?

Download the file `Week3Ex2.txt` from Moodle and consider the data it contains.

We want to investigate

$$H_0 : X \text{ and } Y \text{ are independent.}$$

Tasks

1. Draw a scatterplot of x and y . Describe what you see. Do the variables appear to be related?
 2. Compute the observed Pearson and Spearman correlations. Before doing any test, compare these numbers with what you saw in the plot.
 3. Construct a permutation test of independence using Pearson correlation as the test statistic. Permute y while keeping x fixed and use $B = 4999$ permutations.
 4. Repeat the test using Spearman correlation.
 5. Compare the two p-values with your visual impression from Task 1. What do you conclude?
 6. Is the permutation procedure itself the problem, or is something else responsible for the result?
 7. Find a statistic that better captures the relationship you see in the scatterplot and use it in a permutation test. Explain why you chose it.
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Exercise 3 — Bootstrap uncertainty and the number of replicates

Consider the wind-speed measurements in the `airquality` data included with R.

```
x <- na.omit(airquality$Wind)
```

Tasks

1. First look at the data. Make a histogram and a boxplot of x , and compute a few useful numerical summaries.

Describe the distribution. Comment on its center, spread, skewness and any unusual observations.
2. We are interested in the **population median wind speed**. Compute the sample median.
3. Write a nonparametric bootstrap that resamples the observed wind speeds with replacement and computes the median.
4. Estimate the bootstrap standard error using

$$B \in \{100, 500, 2000, 10000\}.$$

5. Repeat the complete calculation 20 times for each value of B . Store the 20 estimated standard errors.
6. Make a boxplot of these estimates against B . What changes as B increases?

7. Explain the difference between:
 - statistical uncertainty caused by observing only n wind speeds;
 - Monte Carlo error caused by using only B bootstrap replicates.
 8. Why does increasing B not compensate for a small original sample size?
 9. Suppose a bootstrap test needs to distinguish p-values around 0.001. Would $B = 199$ be reasonable? Use the approximate p-value resolution $1/(B + 1)$ in your argument.
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Exercise 4 — Confidence intervals for the median wind speed

Continue with the wind-speed data from Exercise 3. The parameter of interest is the population median wind speed.

Tasks

1. Generate $B = 10000$ nonparametric bootstrap replicates of the median.
2. Construct a 95% **normal bootstrap interval**

$$\hat{\theta} \pm z_{0.975} \hat{\text{se}}_{\text{boot}}.$$

3. Construct a 95% **percentile interval** using the 0.025 and 0.975 quantiles of the bootstrap distribution.
4. Construct a 95% **basic bootstrap interval**

$$\left[2\hat{\theta} - q_{0.975}^*, 2\hat{\theta} - q_{0.025}^* \right].$$

5. **Where does the basic interval come from?** The bootstrap approximates the distribution of

$$\hat{\theta} - \theta$$

by the distribution of

$$\hat{\theta}^* - \hat{\theta}.$$

Starting from the bootstrap quantiles $q_{0.025}^*$ and $q_{0.975}^*$, derive the basic 95% confidence interval above. In particular, explain where the factor $2\hat{\theta}$ comes from.

6. Plot the bootstrap distribution and mark the observed median and the relevant bootstrap quantiles.
7. Compare the three confidence intervals. Are they very similar? Which interval is symmetric around the observed median?
8. A colleague claims that the population median wind speed is **10 mph**. What can you say about this claim based on each of your 95% confidence intervals?
9. Another colleague says:

“Because 10 mph is inside my 95% confidence interval, there is a 95% probability that the true median is between the two endpoints.”

Is this a correct interpretation of a frequentist 95% confidence interval? Explain.

10. Suppose instead that a proposed value lies just outside one of the 95% intervals. Explain the connection with a two-sided hypothesis test at approximately the 5% level.

Exercise 5 — When the ordinary bootstrap fails

Suppose

$$X_1, \dots, X_n \stackrel{iid}{\sim} \text{Uniform}(\theta, 1)$$

with $\theta < 1$, and consider

$$m_n = \min(X_1, \dots, X_n).$$

Generate one sample with $\theta = 0$:

```
set.seed(35)
n <- 30
theta <- 0
x <- runif(n, theta, 1)
m_obs <- min(x)
m_obs
```

```
## [1] 0.01082598
```

Tasks

1. Generate $B = 5000$ ordinary nonparametric bootstrap samples and record their minima m_n^* .
2. Verify numerically that

$$m_n^* \geq m_n$$

for every bootstrap sample.

3. Plot the bootstrap distribution of $m_n^* - m_n$.
4. Generate 5000 new samples from the true distribution $\text{Uniform}(0, 1)$, and record their minima.
5. Compare the bootstrap distribution of $m_n^* - m_n$ with the simulated sampling distribution of $m_n^{\text{new}} - \theta$.
6. Explain why the empirical distribution $\hat{F}_n$ cannot reproduce the relevant uncertainty at the lower endpoint.
7. Contrast this with bootstrapping the sample mean. Why is the minimum more problematic?
8. We have now seen problems for the **minimum** and **maximum**. Find another statistic, not simply the minimum or maximum itself, for which the ordinary nonparametric bootstrap may fail or behave badly.

Explain why you expect a problem. Does the empirical distribution contain the information needed to reproduce the sampling behaviour of your statistic?

Investigate your proposed statistic by simulation if useful.

What to take away

Across all five exercises, the main question is not merely **how to resample**, but

Why is this particular resampling scheme valid for this problem?
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Permutation methods rely on an invariance or exchangeability under the null.

The ordinary iid bootstrap treats the empirical distribution as an estimate of the data-generating distribution.

Both methods can fail when their resampling scheme does not reproduce the structure relevant to the statistic or hypothesis.